Your Personalized Learning Roadmap By PathForge: SQL

1 message

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Learning Overview

This roadmap will take you from a beginner to an advanced SQL practitioner capable of designing, querying, and optimizing relational databases for real-world applications. The 9-week plan is divided into three phases—Foundations, Applied Learning, and Advanced Mastery—each building practical skills through structured daily study, hands-on exercises, and project work. By completion, you will confidently write complex queries, manage databases, and optimize performance, ready to apply SQL in professional environments.

Phase 1: Foundations (Weeks 1-3)

Topics:

- Introduction to Databases & SQL
- Basic SELECT statements
- Filtering data with WHERE
- Sorting and limiting results (ORDER BY, LIMIT)
- Aggregate functions (COUNT, SUM, AVG, MIN, MAX)
- GROUP BY and HAVING clauses
- Basic JOINs (INNER JOIN)
- Data types and simple table creation

Learning Priorities:
Master querying basics and understand relational database concepts. Focus on writing clean, error-free queries.

Daily Study Structure (4 hrs):

2 hrs: Concept learning with examples

1 hr: Guided exercises on topic covered

1 hr: Mini practice tasks and review

Mini Practice Tasks:

Write queries to filter and sort data from sample datasets

Aggregate sales or user data with GROUP BY

Join two tables with INNER JOIN

Milestone Checkpoints:

Write queries using SELECT, WHERE, ORDER BY, and LIMIT on sample datasets

Use aggregate functions and GROUP BY to summarize data

Perform INNER JOIN queries combining two tables

Phase 2: Applied Learning (Weeks 4-6)

Topics:
Advanced JOINs (LEFT, RIGHT, FULL OUTER)

Subqueries and nested queries

Set operations (UNION, INTERSECT, EXCEPT)

Modifying data (INSERT, UPDATE, DELETE)

Creating and altering tables (DDL)

Index basics and their impact on performance

Views and stored procedures introduction

Transaction basics and ACID properties

Hands-on Exercises:

Write multi-join queries combining 3+ tables

Create subqueries for filtering and aggregation

Practice data manipulation commands on test databases

Practical Implementation Tasks:

Design a small normalized database schema

Populate tables and perform CRUD operations

Create views to simplify complex queries

Project Suggestions:

Build a simple inventory management database with queries for reports

Create a user activity tracker with summary views

Milestone Checkpoints:

Write queries using advanced JOINs and subqueries

Perform full CRUD operations and schema modifications

Build and query views, demonstrate transaction usage

Phase 3: Advanced Mastery (Weeks 7-9)

Topics:
Query optimization techniques (EXPLAIN, indexing strategies)

Window functions (ROW_NUMBER, RANK, LEAD, LAG)

Common Table Expressions (CTEs) and recursive queries

Advanced stored procedures and triggers

Database security basics (roles, permissions)

Backup and recovery fundamentals

Working with JSON and semi-structured data in SQL

Performance tuning and best practices

Real-world Applications:

Optimize slow queries using indexing and EXPLAIN plans

Implement business logic with stored procedures and triggers

Secure data access and manage user permissions

Portfolio-level Projects:

Develop a reporting dashboard backend with complex queries and CTEs

Create audit logs using triggers and stored procedures

Milestone Checkpoints:

Analyze and optimize queries using EXPLAIN and indexing

Write advanced queries using window functions and recursive CTEs

Build secure, maintainable stored procedures and triggers

Weekly Study Structure

Day	Activity	Duration	Focus Area
Monday	Learn new concepts	2 hrs	Theory + examples
	Practice exercises	1 hr	Guided query writing
	Mini project/practice task	1 hr	Apply concepts
Tuesday	Review previous day + revision	1 hr	Reinforce learning
	Learn new concepts	2 hrs	Theory + examples
	Practice exercises	1 hr	Query practice
Wednesday	Hands-on project work	2 hrs	Build schema/queries
	Practice problem-solving	1 hr	Focus on weak areas
	Revision	1 hr	Review notes and queries
Thursday	Learn new concepts	2 hrs	Theory + examples
	Practice exercises	1 hr	Query writing
	Mini project/practice task	1 hr	Apply concepts
Friday	Revision + self-assessment	2 hrs	Test knowledge with quizzes
	Practice exercises	1 hr	Focus on complex queries
	Project work	1 hr	Incremental build
Saturday	Deep dive into projects	3 hrs	Build and optimize queries
	Review and documentation	1 hr	Write notes and comments
Sunday	Rest or light revision	Flexible	Avoid burnout, light practice

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Recommended Resources

Official Documentation:

PostgreSQL Official Docs (Highly recommended for examples and advanced topics)

MySQL Reference Manual

YouTube Channels:

freeCodeCamp.org - SQL Full Course

The Net Ninja - SQL Tutorials

Courses:

Khan Academy - Intro to SQL (Free, beginner-friendly)

Mode Analytics SQL Tutorial (Interactive and practical)

Books:

SQL Queries for Mere Mortals by John L. Viescas & Michael J. Hernandez (Practical and clear)

Learning SQL by Alan Beaulieu

Practice Platforms:

LeetCode SQL Problems

HackerRank SQL Practice

SQLZoo

Communities:

Stack Overflow SQL Tag

Reddit r/SQL

DBA Stack Exchange

Capstone Project

Project: Build a Complete E-commerce Order Management System Database
Design normalized tables for users, products, orders, payments, and inventory

Implement complex queries for sales reports, customer analytics, and inventory status

Use stored procedures for order processing and triggers for audit logging

Optimize queries with indexing and analyze performance

Secure data access with roles and permissions

This project demonstrates end-to-end database design, querying, and optimization skills highly valued in industry roles.

Pro Tips

Consistent Short Bursts: Break your 4-hour study into focused 50-minute blocks with 10-minute breaks to maintain concentration.

Active Query Writing: Immediately apply concepts by writing queries instead of just reading theory. Practice beats passive learning.

Use Real Datasets: Practice with real or realistic datasets to understand practical challenges like NULLs, duplicates, and indexing.

Revise Weekly: Spend time revisiting previous topics to strengthen retention and connect concepts.

Track Progress with Milestones: Regularly test yourself against milestones to identify gaps early and adjust learning focus.

PathForge AI Chatbot

Click on the chatbot link to know more about the roadmap and streamline your learning journey.
